

Vision Based Navigation and Control for Autonomous Cassava Farm Robots

Complete Research Technical Report
Phase C Comparative Analysis
2026-09-28

Executive Summary

This report documents the Phase C validation of frozen Phase B evidence. Random Forest has the lowest saved-test error but uses oracle ground-truth-mask features. MobileNetV3 is the lowest-MAE image-only export. The common paired test set contains 42 images.

Dataset and workflow

The usable data contain 802 samples, 3 confirmed semantic classes and geometry-derived navigation targets. Classical ML and ANFIS use oracle feature inputs; direct CNN/transfer models use images. No physical robot claim is made.

Verified comparative results

Metrics were recomputed from saved actual/predicted pairs. Bootstrap confidence intervals use 5,000 image-level resamples; paired Wilcoxon tests use the 42 shared image IDs and Holm correction.

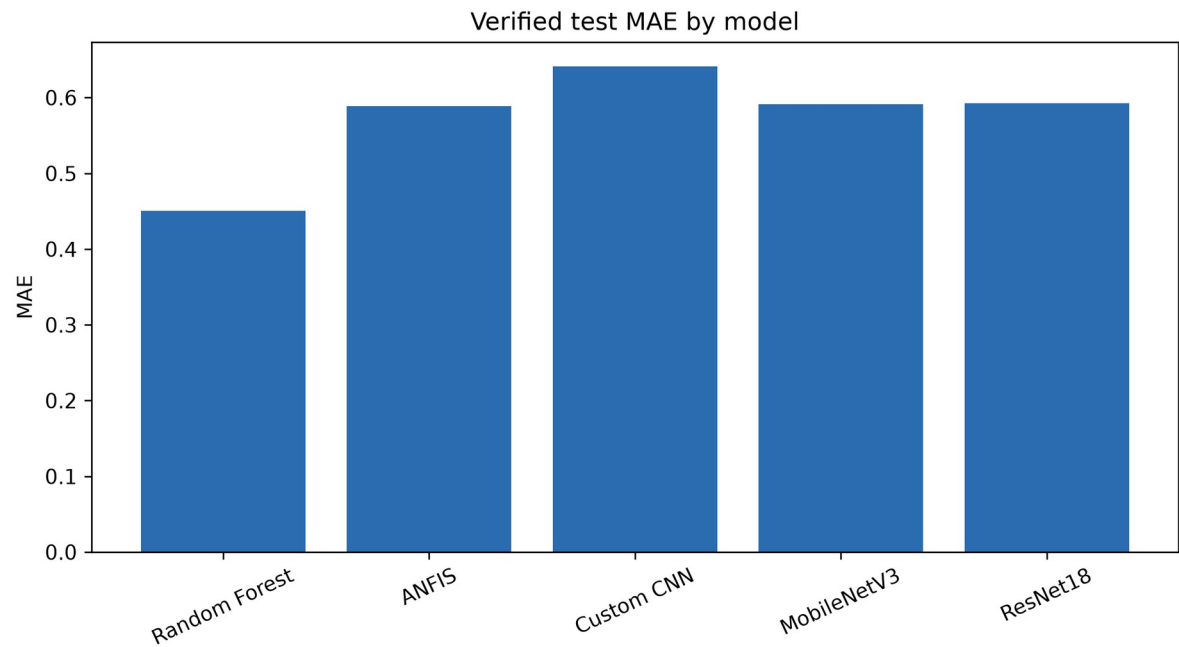


Figure - Verified test MAE by model

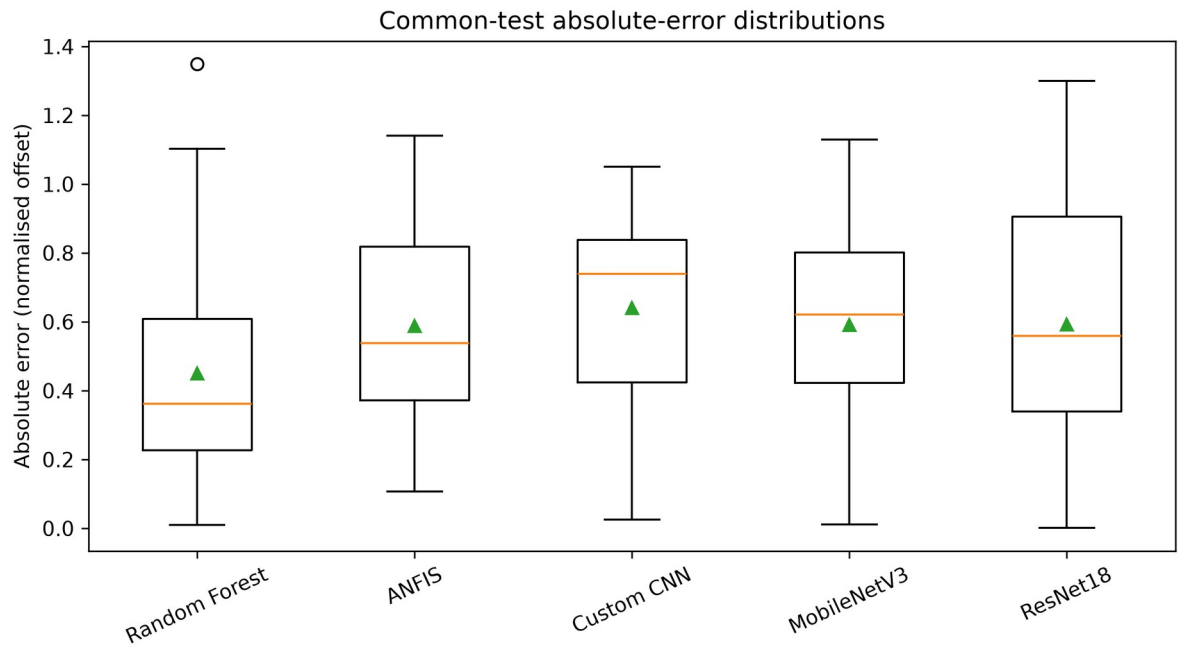


Figure - Paired absolute-error distribution

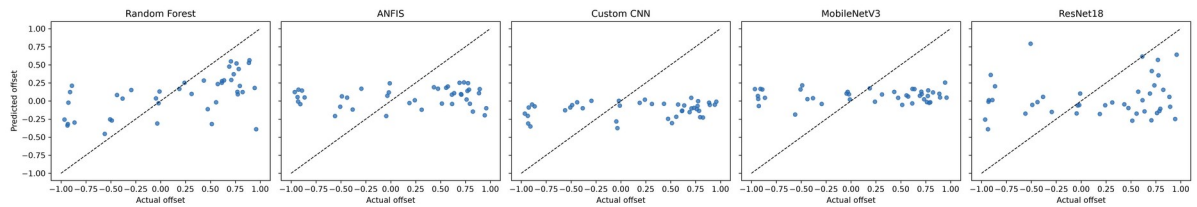


Figure - Actual versus predicted common-test offsets

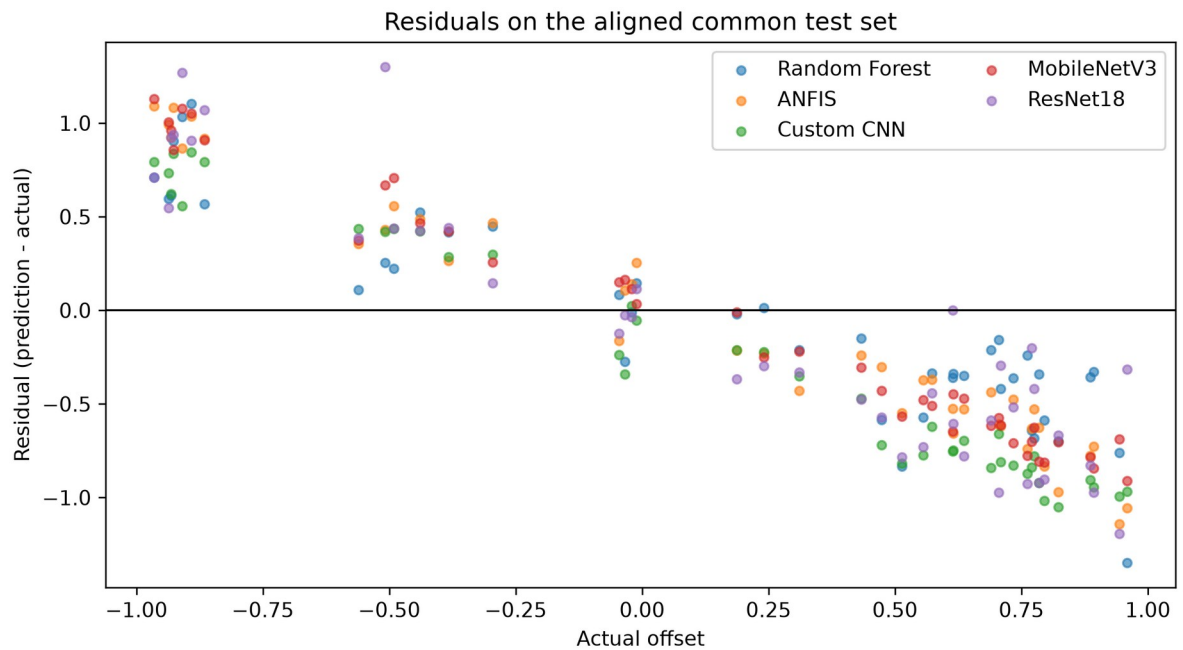


Figure - Residual comparison

Interpretation and limitations

The oracle/image-only distinction prevents a direct deployment ranking. Targets are not measured steering commands, and segmentation execution, predicted-mask features, edge testing and robot validation remain unavailable.

Verified regression table

Model	Feature source	N	MAE	RMSE
Random Forest	oracle ground-truth masks	42	0.4505 698 017 042 929	0.5420 931 192 289 319
ANFIS	oracle ground-truth masks	42	0.5886 896 949 761 904	0.6565 839 887 087 348
Custom CNN	images	42	0.6414 214 591 595 238	0.6965 784 109 611 107
MobileNetV3	images	42	0.5915 980 909 523 809	0.6582 767 348 218 261
ResNet18	images	42	0.592889 465 952 381	0.685875 004 015 314

Appendix and reproducibility

See the accompanying Markdown report, figure catalogue, table catalogue, objective traceability and reproducibility guide for complete traceability.